Sweatshop on P-Chess: lane `core`, afternoon

The UCI bridge, a game against Maia and the clocks (BOT-002, BOT-004, CLOCK-001); Opus 5.5 lineup

Foreman, session `sweatshop/core/2026-10-24-1535`

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Summary

Lane `core` ran in the clone `.artifacts-build/lanes/core`, from `main` at `7813c75` (PR #5 merged, so BOT-001 and its engine fetch script were on the base). Every open ticket of the three lanes waited on one chain here: BOT-002 → BOT-004 → CLOCK-001. Two launches, both with `-Models 'stage 2 opus-5.5 high, hard opus-5.5 xhigh, stage 3 opus-5.5 xhigh'` (Bruno's choice; the same models as the `night` binding, logged by the driver as `ad hoc`): 15:35 (pid 18936, ended 16:11 "Nothing runnable" after the driver parked a valid BOT-004 handoff) and 16:12 (pid 9808, stopped with `STOP` at 17:55, once CLOCK-001 was merged, as the supervising session asked). Before launch the foreman fetched `engine/` into the clone with `scripts/fetch-engine.ps1` (every hash verified, gitignored), so the live Lc0 tests ran instead of being skipped.

- **Produced: 3 tickets approved and merged on the session** PR #6: BOT-002, BOT-004, CLOCK-001, all `Review: agent`, `Verdict: Approve`, no *Needs your call.* BOT-004 and CLOCK-001 close with `Acceptance: pending` (manual checks, release impact).
- **Cost: $24.89** at API list price over 8 stages, all priced: `opus-5.5 xhigh` $17.09 (hard stage 2 and every review), `opus-5.5 high` $7.79. No Codex model ran a stage (Codex quota: `-`); one `gpt-6-astra medium` call judged the reopen (section Reopen attribution).
- **What went wrong:** BOT-004's 50 ms latency criterion. The first review measured 74–106 ms in the windowed app and reopened; the retry found the machine saturated by three lanes and another project (Lc0 alone 247–813 ms per move). The proxy turned the target into manual acceptance and Bruno deferred the measurement to an idle machine. The reopen cost $4.54 (review plus retry).
- **Provisional verdict:** Opus 5.5 high implementing under Opus 5.5 xhigh review delivered the chain in 2h20 with one reopen, whose cause was the environment more than the code. The live engine paid for itself at once: it found a real shutdown bug in BOT-002 that the rebuilt fixture had hidden.

What was done

BOT-002: the game can now talk to Lc0. `engine/uci.gd` starts Lc0 once, reads it on worker threads, switches Maia nets, and returns Maia's policy (a probability per legal move) on the main thread. The reviewed work of the night run was cherry-picked (its merge had been reverted for want of BOT-001), and this time the parser's fixture was recorded from the real Lc0 v0.32.1 and the level-switch test ran live. The live run caught a bug: the reader threads ran a method of the bridge node, which in debug builds locked the node, so `free()` failed and Lc0 was never stopped; the method became `static`. The recording also settled that Lc0's default `PolicyTemperature` is 1.359.

BOT-004: a player can pick a colour and a Maia level (1100–1900) on a new setup screen and play a full game against her. `bot/maia_player.gd` drops illegal policy entries, samples with BOT-003's move picker, and logs Maia's reply latency, printing the median at game end; the board ignores input while Maia is to move. Reopened once: criterion 4 (median under 50 ms) failed in the

windowed app with a human-like pause (74.4 and 105.5 ms), with the machine under heavy load. Not an implementation fix: the proxy rewrote criterion 4 to "record and print the median" and Bruno deferred the 50 ms check to an idle machine.

CLOCK-001: games are timed. `core/clock.gd` counts milliseconds, leaves each side's first move untimed and without increment (as Lichess does), plays the low-time warning once, and flags at zero; the setup screen offers 1+0, 3+2 and 5+0, and a flag ends the game through CORE-004's rule. A proxy decision added criterion 8 and `ui/board_view.gd`: a piece dropped after the game ended snaps back instead of adding a move. The review's three mutants each failed their own test.

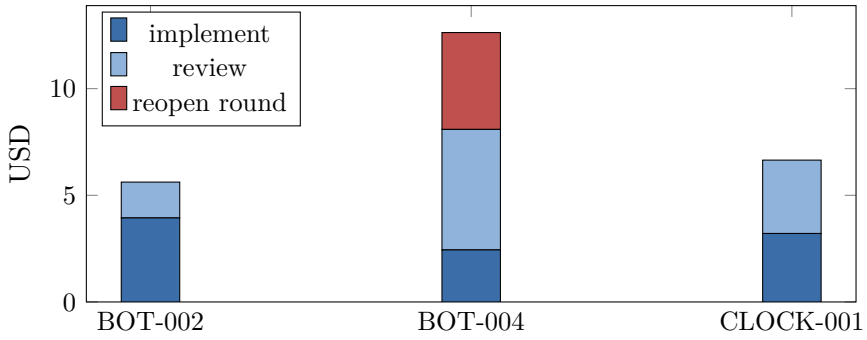
Timeline

15:35 Clone pulled to `7813c75`; `engine/` fetched. Session opened locally (`43fe465`); BOT-002 restored to `to-implement` (`c99686f`), old branch kept as `recovery/20261024-153504/BOT-002`. Launch, pid 18936.
15:47 BOT-002 implement (hard): to review, fixture recorded, c7 live, 86/0.
15:56 BOT-002 review: merged (`5143b81`).
16:10 BOT-004 implement: valid handoff (92/0) parked as "missing Gate evidence receipt".
16:11 Driver ends; PR #6 opened.
16:14 Foreman: receipt line fixed on the ticket branch, park `d44ea41` reverted (`1f45929`). Relaunch, pid 9808.
16:28 BOT-004 review: reopened, criterion 4.
16:45 Bruno (relayed): defer the latency check to an idle machine; recorded on the session (`3777aad`).
16:53 BOT-004 retry: to review, no code change, criterion 4 rewritten by the proxy.
17:12 BOT-004 review: merged (`34e432c`).
17:41 CLOCK-001 implement: to review, 102/0. `STOP` created as the review starts.
17:55 CLOCK-001 review: merged (`c160ef0`); driver stops; PR #6 updated (3 tickets).

Per stage

When	Ticket	Stage	Model	Min	In	Cache	Out	Cost	Outcome
15:47	BOT-002	impl	opus xhigh	12	7.7M	97%	52k	3.94	to-review
15:56	BOT-002	review	opus xhigh	9	2.7M	95%	47k	1.68	merged
16:10	BOT-004	impl	opus high	14	4.5M	98%	38k	2.44	to-review*
16:28	BOT-004	review	opus xhigh	15	4.5M	96%	52k	2.40	reopened
16:53	BOT-004	impl	opus high	23	3.4M	96%	33k	2.14	to-review
17:12	BOT-004	review	opus xhigh	19	11.3M	98%	86k	5.65	merged
17:41	CLOCK-001	impl	opus high	29	5.9M	97%	47k	3.21	to-review
17:55	CLOCK-001	review	opus xhigh	13	6.4M	96%	72k	3.44	merged
				134	46.4M		427k	24.89	

* The run-log row reads `failed (incomplete handoff), blocked`. The handoff was valid; the driver did not read its receipt line (Driver issues, 1). The row is left as logged.



The red segment is BOT-004's first review and its retry ($4.54), spent on a criterion the machine could not measure. BOT-004's closing review was the costliest stage (11.3M input, $5.65): it resolved four rebase conflicts with Bruno's deferral, moved BOT-002's open notes to other tickets and rebuilt a gate receipt by hand. Input is 95–98% cache reads.

Comparison with previous runs

From `run-log.md`: the night `core` run, same lineup family, cost $7.20 per approved ticket; lane `mvp` $2.69 (small tickets, day lineup); this run $8.30 ($24.89 over 3). These were the heaviest tickets so far (a process bridge with threads, a first playable bot screen, a clock wired through the UI), and BOT-004 alone is 54% of the cost. Stage 2 took 12–29 min here (median 18.5) against a 6 min median for extttopus-5.5 high overall.

Observations per model

- **opus-5.5 xhigh, hard stage 2 (BOT-002)**: reused the reviewed work as told, recorded the fixture from the live engine, and found the shutdown bug from a failing live test, with the before and after kept as proof.
- **opus-5.5 high, stage 2 (BOT-004 twice, CLOCK-001)**: every handoff disclosed what it had not checked (manual games, drag smoothness, the one-frame flag race). On the BOT-004 retry it measured, found the machine saturated, and asked the proxy instead of tuning `engine/uci.gd` outside its files. Its first BOT-004 handoff wrote the receipt line as a bullet with a note, which the driver cannot read.
- **opus-5.5 xhigh, review**: three merges and one reopen; no code authored in review, so no round-0 rows. It measured criterion 4 in the real windowed app rather than trusting the headless test, ran mutants on CLOCK-001, and turned every open note into a ticket or a destination (BOT-008, CLOCK-002, notes on BOT-005, BOT-007, HIST-002). Two habits cost elsewhere: notes written into other lanes' tickets, and a gate receipt written by hand when the helper was denied.

Driver and runtime issues

1. **`- Gate evidence:` is invisible to the driver.** `Field()` only reads a line that starts with `Gate evidence:`; a bullet, or a trailing note after the path, parks a valid handoff. Fourth time today (CORE-001, CORE-005 in the night run, BOT-004 here). It also mischarges the scoreboard: the next review is attributed to implementer `?`.
2. **`gate-evidence.ps1` denied in stage sessions.** BOT-002's stage 2 got it through only in one exact Bash form (a proxy decision); BOT-004's review could not run it and wrote the receipt by hand from a manual gate run. The receipt matched, but a hand-written receipt is not the evidence the check was built for.
3. **A merged-and-reverted branch cannot be re-reviewed.** BOT-002's commits were ancestors of `main` through the reverted merge, so restoring `to-review` would have merged nothing. The foreman restored `to-implement` with a cherry-pick instruction and quoted the stale record so the driver would not read its old receipt, verdict and acceptance lines.
4. **Shared machine.** A latency criterion ran with three lanes and another project on the same PC; neither the driver nor the stage can tell code cost from load.

Recommendations

1. Make the driver's `Field()` accept an optional leading `-` and stop at whitespace after a path, or have ticket-flow's handoff template show the exact line. Rests on four parked handoffs today; scoreboard row `?` → `opus-5.5 xhigh` (3 reviews) exists only because of it.

2. Allow the gate helper's command form in the stage sessions' permissions (Bruno's call), so no review writes a receipt by hand.
3. Write timing criteria as manual acceptance on an idle machine from the start, or run them only when no other lane runs. BOT-004's reopen round cost $4.54.
4. In ticket-flow, route a review's notes for another lane's ticket to its own ticket or the base, not into the other lane's folder: BOT-005 and BOT-007 will conflict when the bot and mvp lanes sync.
5. Keep `opus-5.5 high` → `opus-5.5 xhigh` for the core chain: 8 reviews, 2 reopened, S2 rate 25% (an anecdote under 10 reviews), plus BOT-004's reopen mischarged to `?`.

Human debt and proxy decisions

No `Débito humano:` lines this run. Bruno decided twice, relayed by the supervising session: unblock BOT-002 after BOT-001 merged; defer BOT-004's latency measurement until the machine is idle ("deixa a parte do tempo de resposta pra depois"). `Proxy decided` lines, all `(opus)`, spawned by the stages:

- **BOT-002**: retry `gate-evidence.ps1` once as a single `powershell -NoProfile -File` Bash command; never hand in work without a receipt. (The night run's option A, a rebuilt fixture, is superseded by the recording.)
- **BOT-004**: criterion 4 keeps the 50 ms target as M3 product acceptance, measured in a manual game on an idle machine; the loop checks only that latency is recorded and its median logged; `engine/uci.gd` stays out of Primary files.
- **BOT-004**: `PolicyTemperature` stays Bruno's question, carried by notes on BOT-005 and BOT-007; the missing Lc0-stop detection becomes ticket BOT-008.
- **BOT-004**: hand-run the gate helper's steps for a fresh receipt on `0ad3d14`, the helper being denied by permissions.
- **CLOCK-001**: a side's first move is untimed and adds no increment, as on Lichess (criterion 2).
- **CLOCK-001**: `ui/board_view.gd` (`_commit`) and its test join Primary files with criterion 8; a drop after the game ends snaps back.

Reopen attribution

One reopened review; judged by `gpt-6-astra medium`, read-only, since the reviewer is Opus. No merging review authored a fix, so no round-0 rows. PR #6 has no review findings yet; no session-review, release-audit or post-release discovery.

Ticket	Rnd	Finding	Label	Why
BOT-004	1	Windowed median 74.4 / 105.5 ms against criterion 4's 50 ms	S2-explícito/ código	The original criterion explicitly requires under 50 ms; the measurements do not isolate code cost from machine load. Review miss: no. Churn: no.

Totals: S2-explícito/código 1. The foreman notes that the cause was most likely load, not code (the retry measured Lc0 alone at 247–813 ms), and that the criterion was later rewritten by the proxy; the label stands as returned. The scoreboard charges this round to `?`, not to `opus-5.5 high` (Driver issues, 1).

Scoreboard

`scoreboard.ps1` over P-Chess (main checkout, log merged and copied back), ClassyCall, EasyMath, PhysiSyst, SIGAA-ME, SynchroNice and slip. Rows that moved since the `mvp` report: `opus-5.5 high` → `opus-5.5 xhigh` 6 → 8 reviews, still 2 reopened (33% → 25%); `opus-5.5 xhigh` → `opus-5.5`

`xhigh` 5 → 7, 0 reopened; a new row `?` → `opus-5.5 xhigh` 3 reviews, 1 reopened, 1 unclassified (BOT-004's round, mischarged). All Opus pairs remain under 10 reviews: anecdotes. The attribution and intervention rows of this run are committed on `main` with this report; the printout was taken from the main checkout before that commit, so they are not in it.

Implementer	Reviewer	Reviews	Reopened	Rate	S2	S2 rate
gpt-6.1-sol high	gpt-6.1-sol xhigh	89	12	13%	11	12%
gpt-6.1-sol high	gpt-6.1-sol max	94	21	22%	20	21%
opus-5.5 high	opus-5.5 xhigh	8	2	25%	2	25%
opus-5.5 xhigh	opus-5.5 xhigh	7	0	0%	0	0%
?	opus-5.5 xhigh	3	1	33%	0	≥0%
sonnet-5.5 high	opus-5.5 xhigh	3	0	0%	0	0%

Implementers: `opus-5.5 high` 14 runs, 10 to review, 4 failed, median 6 min, $24.96; `opus-5.5 xhigh` 19 runs, 14 to review, 5 failed, median 12 min, $90.74. Interventions across repos: 22 human rescues, 48 automatic recoveries, 6 routine approvals, 30 `gpt-6-astra high, fallback`, 15 Opus and 12 supervisor proxy decisions, 37 unknown. Runtime completion to process exit: 854 s over 609 measured stages, 756 unmeasured. Full printout: the foreman's scratch output.

Pending for the human

1. Merge PR #6 (BOT-002, BOT-004, CLOCK-001); then the bot lane (BOT-005, BOT-006) and later mvp (BOT-007, CI-002) can start, and the other open lanes sync.
2. BOT-004: measure Maia's latency with the machine idle (a windowed game against maia-1500 as each colour; median under 50 ms). Over 50 ms: a ticket with `engine/uci.gd` in Primary files for Lc0's thread options.
3. CLOCK-001: lose a 1+0 game on time; win a game on time against Maia.
4. `PolicyTemperature`: pick the value (Lc0's default is 1.359); BOT-005 and BOT-007 carry the question.
5. Stage 1: register BOT-008 in the plan; choose CLOCK-002's test approach (it is `blocked`); fold the HIST-002 time-spent note (first move shows 0 s).